

NLP Assignment – Project Validation Submission

CCS3356 - Natural Language Processing



Group members :

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Group Information

Group Number: 17

Project Title: Cross-Platform Spam Detector

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Section 1 – Project Overview

1. What NLP problem/application are you attempting to solve?

We are developing a **Cross-Platform Spam Detector** using Natural Language Processing (NLP). The system will automatically classify the message as **spam or not spam**. Spam content is typically found on multiple platforms, such as **SMS, emails, and YouTube comments**. Manually finding spam messages can be tedious and time-consuming for users. The proposed system will automatically identify spam by analysing the textual content of messages and comments.

The text data will be preprocessed and cleaned using NLP techniques before classification. We will train Machine Learning and Deep Learning models to identify spam patterns in different text types.

The system will be trained on labelled datasets, including spam and legitimate messages. Once trained, **the models will be able to correctly classify new messages**. The best models will be included in a user-friendly application.

In this project, we show how NLP can be used to improve security in online communication and minimise unwanted content across different digital platforms.

2. Why is this problem important?

Spam is a common problem in **email, SMS, and YouTube comments**. It can consume time, spread scams, and risk user privacy. This system is designed for students, businesses, content creators and general users. **It auto-detects suspicious messages and helps users to be safe and save time**. The solution also allows organizations to more efficiently manage large amounts of content online.

3. What will be the final output of your application?

- A simple and easy-to-use web interface will be given.
- The NLP model will process the user's text and categorise it as
 - **Spam (Not secure)**
 - **Ham (Safe)**
- Show classification results in a clear way with colour indicators:
 - **Red warning alert** for spam messages.
 - **Green success box** for safe messages
- The system will show **an AI confidence score**, as a percentage (e.g., 96% Probability of Spam), to indicate the level of confidence of the model.
- Additional security-related features will be included:
 - ✓ Automatic detection of links and URLs in the text.
 - ✓ Warning messages for detected links or URLs.
 - ✓ Highlighting of suspicious keywords such as:
 - **FREE**
 - **Winner**
 - **Crypto**
 - **Lottery**
 - **Click Here / Click Link**
 - **bit.ly, goo.gl, http://, https://**
- Explanation of why a message may be classified as spam.
- Users will be able to select the platform from which the message originates:
 - **Email**
 - **SMS**
 - **YouTube**
- Based on the selected platform, the system will provide simple **safety tips**.
- These features will help users:
 - Identify spam messages more effectively.
 - Understand the model's prediction.
 - Improve their online safety and awareness.

Section 2 – Dataset Information

4. Dataset Details

- **Dataset Name:** Combined Cross-Platform Spam Dataset
- **Dataset Source (URL):**
 - **Source 1 (SMS):** <https://www.kaggle.com/datasets/hamnawaseem112222222/sms-spam-collection-5572-labeled-sms-messages>
 - **Source 2 (Email):** <https://www.kaggle.com/datasets/venkateshch22384/spam-and-ham-email-dataset>
 - **Source 3 (YouTube):** <https://www.kaggle.com/datasets/fatihblg/youtube-spam-collection>
- **Number of Records:** 11,911 total combined rows (**SMS: 5,572 + Email: 4,845 + YouTube: 1,494**)
- **Number of Classes:** 2 Classes
 - Spam
 - Ham (Not Spam)

5. Why is this dataset suitable for your project?

The combination of data sets is extremely relevant in this case, due to the inclusion of realistic data in three different communication streams: **SMS, emails, and YouTube comments**. Unlike using just one source of data, the combined format will enable our models to understand a **variety of text formats**, from professional **emails to mobile slangs and social media comments**. The presence of almost 12,000 data points will offer an adequate baseline for comparing the performance of our innovative ML and DL models.

6. What challenges exist within the dataset?

- **Imbalanced Classes:** The classes in the dataset are imbalanced in terms of the number of records in each class, since normal records (Ham) are more than spam records (SMS dataset contains 87% Hams and just 13% Spams). It might lead to model biasness towards a particular class.
- **Noisy Text:** The text in the data consists of high-level noise like use of internet slang language ("u", "plz"), spelling errors, emoticons, character manipulation for evasion of filtering process ("m0ney", "FR3E").
- **Ethical Concerns:** Considering that the source of our data is genuine users' message and comments on social media platforms, there is a slight level of privacy concern. It should always be ensured that no data leaks are done to protect personal information.

Section 3 – Individual NLP Pipeline

Member 01

Step	Justification
Data Collection	Combine SMS, Email, and YouTube spam datasets, combined into one dataset.
Text Cleaning	Remove URLs, punctuation, numbers, special characters, and unnecessary spaces.
Text Normalization	Convert text to lowercase and standardize word formats.
Tokenization	Split text into meaningful words or tokens.
Stop Word Removal	Remove common words that do not contribute significantly to classification (e.g., 'is', 'the').
Lemmatization	Convert words to their base forms to reduce vocabulary size.
TF-IDF Feature Extraction	Convert text into numerical vectors for Machine Learning models.
Model Training & Evaluation	Train and evaluate spam classification models.

ML Model Selected

- Logistic Regression

Reason: Logistic Regression is a simple, fast and very effective algorithm for text classification tasks. It works well with TF-IDF features and serves as a good baseline model for spam detection.

DL Model Selected

- LSTM (Long Short-Term Memory)

Reason: LSTM can capture sequential relationships and contextual information in text. It is a popular method in NLP tasks, and it performs impressively in spam pattern detection in different types of messages.

What contribution are you responsible for in the project?

- Dataset integration
- Text preprocessing
- TF-IDF implementation
- Logistic Regression implementation
- LSTM implementation
- Performance evaluation

Member 02

Step	Justification
Exploratory Data Analysis (EDA)	Analyze dataset distribution and class balance.
Text Cleaning	Remove noisy characters, links, and irrelevant content.
Tokenization	Break text into tokens for processing.
Stemming	Reduce words to their root forms.
Stop Word Removal	Remove frequently occurring but less informative words.
Count Vectorization	Convert text into numerical representations using word frequencies.
Model Training	Train Machine Learning and Deep Learning models.
Hyperparameter Tuning	Improve model performance and generalization.

ML Model Selected

- Support Vector Machine (SVM)

Reason: SVM is particularly useful for text classification because it works efficiently with high-dimensional feature spaces. SVM typically performs very well in spam filtering applications

DL Model Selected

- GRU (Gated Recurrent Unit)

Reason: GRU is the less complex version of LSTM that uses fewer computational resources but manages to capture relevant sequential information.

What contribution are you responsible for in the project?

- Exploratory Data Analysis
- Count Vectorizer implementation
- SVM model development
- GRU model development
- Hyperparameter tuning
- Model comparison

Member 03

Step	Justification
Data Validation	Verify dataset quality and consistency.
Advanced Text Preprocessing	Prepare data for transformer-based models.
Tokenization using BERT Tokenizer	Convert text into tokens suitable for contextual embeddings.
Word Embedding Generation	Capture semantic meaning and context.
Feature Selection & Optimization	Improve model efficiency and reduce noise.
Model Training	Train advanced ML and DL models.
Performance Evaluation	Assess model effectiveness using evaluation metrics.
Deployment Preparation	Integrate the best model into the final application.

ML Model Selected

- Random Forest

Reason: Random Forest is robust to noise, it performs well with complex feature interaction and has a low possibility of overfitting with ensemble learning techniques.

DL Model Selected

- BERT (Bidirectional Encoder Representations from Transformers)

Reason: BERT is a state-of-the-art NLP technique which understands the context based on left and right of the sentence. This is expected to give best results for cross platform spam detection.

What contribution are you responsible for in the project?

- BERT preprocessing and implementation
- Random Forest implementation
- Model evaluation and comparison
- Web application development
- Final integration and deployment

Section 4 – Model Comparison Plan

7. Summarize all selected models.

Member	ML Model	DL Model
Member 1	Logistic Regression	LSTM
Member 2	Support Vector Machine (SVM)	GRU
Member 3	Random Forest	BERT

8. What evaluation metrics will be used to select the best ML and DL models?

The performance of all Machine Learning and Deep Learning models will be looked at using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

We need to use these metrics because Machine Learning and Deep Learning models need to be tested in ways.

Accuracy will show us how many messages are correctly classified. However, since some datasets for spam detection have more of one type of message than the other, we cannot just use accuracy.

Precision will tell us how many messages that are predicted as spam are actually spam. This is important for Machine Learning and Deep Learning models because we do not want to label messages that are not spam as spam as this can be bad for users.

Recall will show us how many actual spam messages are found by the Machine Learning and Deep Learning models. If we have a recall, we can reduce the risk of missing spam messages.

F1-Score gives us a balance of Precision and Recall. This makes it a good way to evaluate systems that detect spam because we care about both messages that're not spam but are labeled as spam and spam messages that are not found.

ROC-AUC will show us how well the Machine Learning and Deep Learning models can tell the difference between spam and non-spam messages and it does this across thresholds for classification.

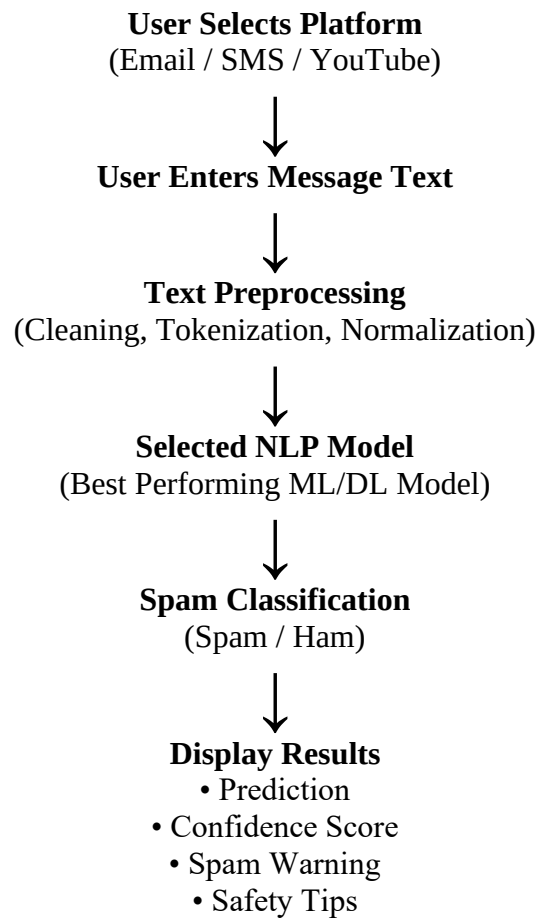
The Machine Learning and Deep Learning model that does the best will be chosen based on how it does, with all of these metrics not just one of them.

Section 5 – Final Application Plan

9. What type of application will be developed?

- Web Application

10. Briefly describe how the application will work.



Section 6 – Workload & Git Management

11. Work Distribution

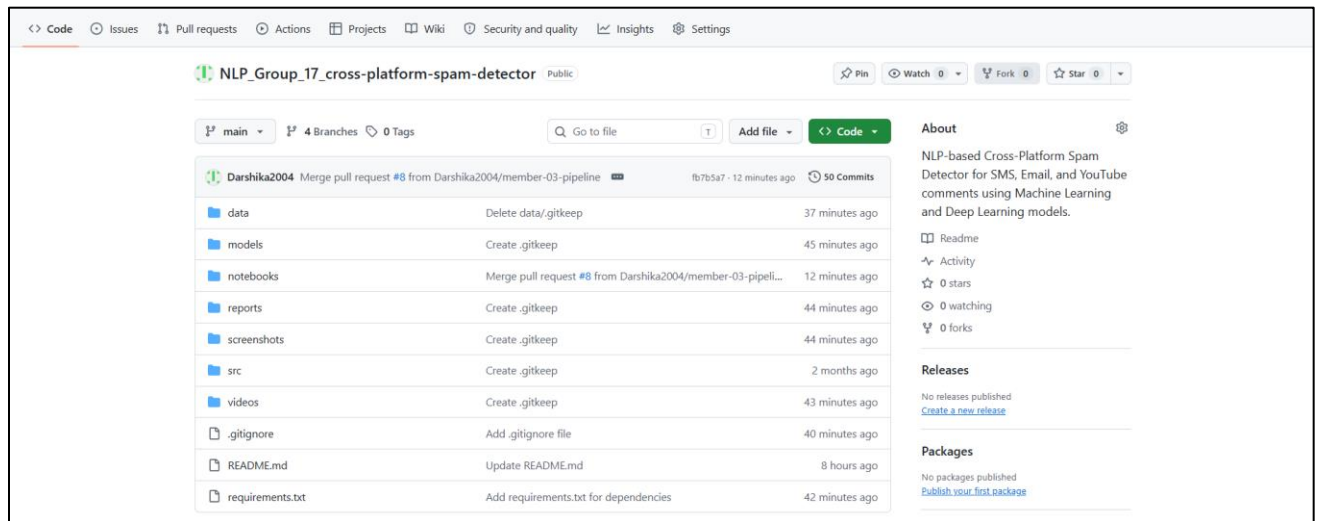
Task	Responsible Member/Members
Dataset Identification	All Members
Data Preprocessing	• P.D.M. Sathsarani (SMS Dataset) • M.K.D.S. Amarasinghe (Email Dataset) • G. Darshika Nuwangani (YouTube Dataset)
ML Model Development	• P.D.M. Sathsarani (Logistic Regression) • M.K.D.S. Amarasinghe (SVM) • G. Darshika Nuwangani (Random Forest)
DL Model Development	• P.D.M. Sathsarani (LSTM) • M.K.D.S. Amarasinghe (GRU) • G. Darshika Nuwangani (BERT)
Model Evaluation & Comparison	All Members (Evaluating individual models)
Application Development	All Members
Report Writing	All Members
Presentation Preparation	All Members

12. Git Repository

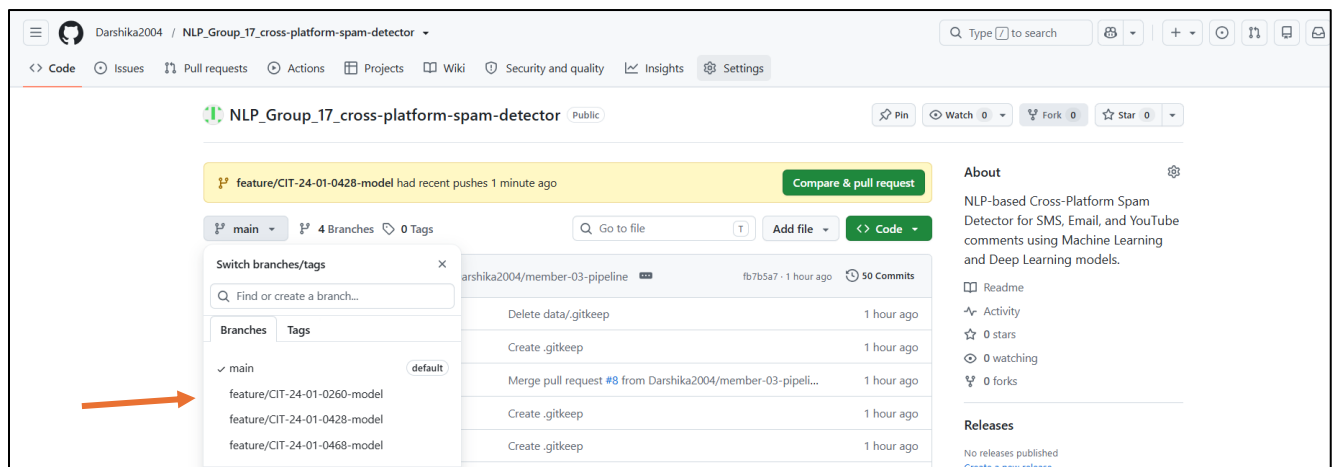
- Repository Link: https://github.com/Darshika2004/NLP_Group_17_cross-platform-spam-detector

screenshots of

- **Repository structure**



- **Branches**



Section 7 – Ethics & Responsible AI

13. Identify potential biases and Ethical problems in the project.

Collecting data on certain platforms such as SMS, E-Mail, and YouTube would lead to the introduction of bias in the dataset, resulting in the performance of the model depending on the kind of message used. It could be possible for some styles of writing, keywords or ways of communication to be considered spams due to their association with spam emails. This would bring about ethical implications relating to the appropriate use and privacy of users' data.

14. Could your system produce harmful or misleading outputs?

Yes. Since Machine Learning and Deep Learning models are not completely accurate, the system may sometimes give wrong predictions.

Examples:

- **"Congratulations! You have won Rs. 500,000. Click the link below."**
 - In the case where the content of the spam email contains new words or structures which are not in the training data set, then there is a chance that the system will fail to recognize it as spam.
- **"Your bank account statement is ready. Please log in to view your account details."**
 - Since the email contains words like "account," "log in," and a link, there is a chance that the system will wrongly predict it as a spam when actually it is not a spam.
- YouTube comments like **"Check out my channel for more tutorials"** will be flagged as spams, but they may actually not be spams.

These misclassification errors can cost users some important emails and even harm them. Hence, it is advisable to use this system only as an aid to decision making.

15. How will you reduce these risks you identified in Questions 13 and 14?

- Ensure you use an unbiased data set from various sources (SMS, Email, YouTube).
- Perform data cleansing to eliminate any data that might distort the analysis process.
- Anonymize the data used for protection purposes.
- Test the model on a broad dataset to enhance accuracy and minimize misclassifications.
- Specify a confidence level for uncertain outputs to be manually examined.
- Keep updating and retraining the model with new data to improve its functionality.
- Let your users know that the system only provides assistance but is not 100 % accurate.

16. What limitations should future users be aware of?

- Not 100% accurate, the system may also give false positives or false negatives.
- The performance will depend on the training set.
- It might have some problems dealing with unknown spam patterns.
- Some emails may be incorrectly identified because of their tone.
- It may not perform well with several languages or even slang.
- It is supposed to serve as a support tool and not as the final decision maker.

Declaration

We confirm that the information provided accurately reflects the current status and planned direction of the project.

Member 1 Signature: 

Member 2 Signature: 

Member 3 Signature: 

Date: 16/06/2026